

Lab 1

Title: Implementing the iterative algorithms in C programming

1. Objective

The objective of this laboratory work is to implement and study the following iterative algorithms:

1. Finding GCD (Greatest Common Divisor)
2. Displaying Fibonacci Sequence
3. Linear Search
4. Bubble Sort
5. Selection Sort
6. Insertion Sort

2. Theory

Algorithms are step-by-step procedures used to solve computational problems. Iterative algorithms use loops instead of recursion, making them efficient in terms of memory usage and easier to understand.

- **GCD** is the largest number that divides two numbers without leaving a remainder.
- **Fibonacci sequence** is a series where each number is the sum of the two preceding numbers.
- **Linear search** is a simple searching technique that checks each element sequentially.
- **Sorting algorithms** arrange data in a specific order (ascending or descending) to make searching and processing easier.

3. Algorithm

a) Algorithm to Find GCD (Iterative)

1. Read two integers a and b
2. While b is not equal to 0
 - o temp = b
 - o b = a mod b
 - o a = temp
3. Display a as GCD

b) Algorithm to Display Fibonacci Sequence

1. Read number of terms n
2. Initialize first = 0, second = 1
3. Repeat n times
 - o Print next term
 - o Update first and second

c) Algorithm for Linear Search

1. Read number of elements n
2. Read array elements
3. Read search key
4. Compare key with each element
5. If found, display position
6. Else display not found

d) Algorithm for Bubble Sort

1. Read array size n
2. Read array elements
3. Repeat n-1 times
 - o Compare adjacent elements
 - o Swap if they are in wrong order
4. Display sorted array

e) Algorithm for Selection Sort

1. Read array size n
2. Read array elements
3. Select minimum element from unsorted part
4. Swap with first unsorted position
5. Repeat until array is sorted

f) Algorithm for Insertion Sort

1. Read array size n
2. Read array elements
3. Take one element and insert it at correct position
4. Repeat for all element

4. Source Code

5. Result

All the given algorithms were successfully implemented using iterative techniques. The programs produced correct outputs for various test inputs.

6. Conclusion

From this laboratory work, we learned how fundamental algorithms such as GCD, Fibonacci, searching, and sorting can be implemented using iteration. These algorithms form the backbone of data processing and problem-solving in computer science. Understanding their working improves logical thinking and algorithmic efficiency.